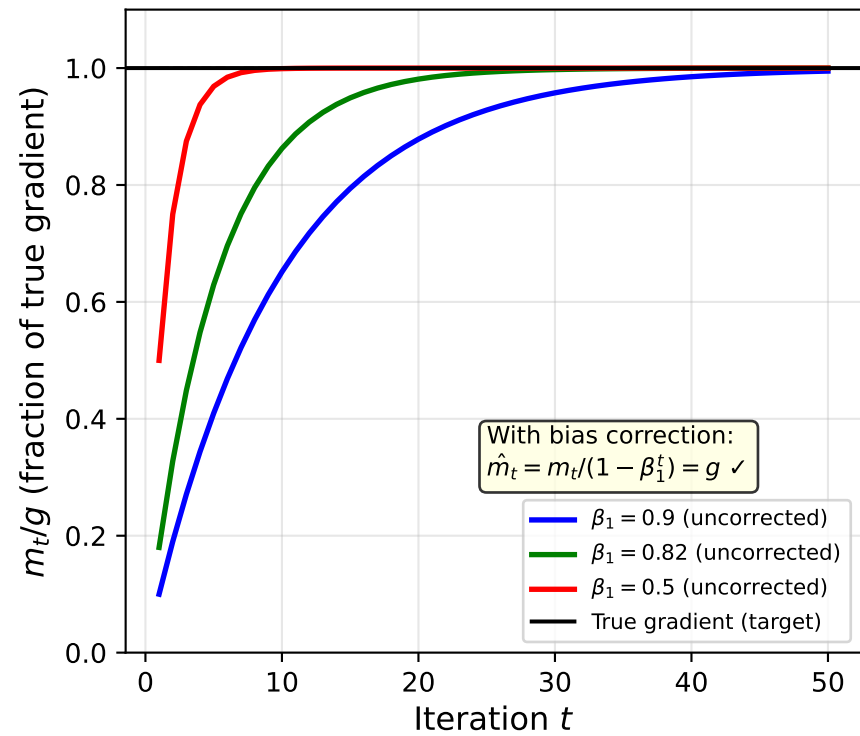
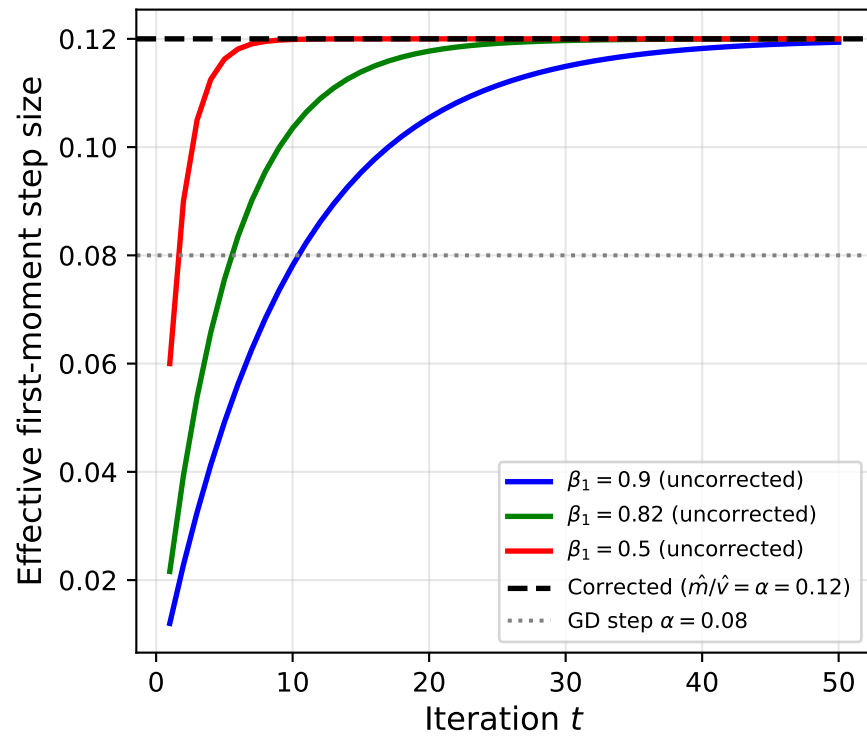


First Moment Without Bias Correction
($m_t = \beta_1 m_{t-1} + (1 - \beta_1)g$)



Without Bias Correction: Early Iterations
Have Too-Small Steps



Bias Correction Factor
(amplifies early estimates)

